PY1(a) – Calculate Factorial

1(a) **Write a program to calculate the factorial of the given number using for loop (Don't use built-in function factorial).**

Aim: To calculate the factorial of the given number using for loop.

Coding:

```
num = int(input('Enter a Number: '))
fact = 1
for i in range(1,num+1):
        fact = fact * i
print("Factorial of ", num, " is ", fact)
```

Output:

```
Enter a Number: 5
Factorial of  5  is  120
```

Result:

Thus the Python program to calculate factorial has been done and the output is verified.

PY1 (b) - Sum of Series

1(b) **Write a program to sum the series $1^1/1 + 2^2/2 + 3^3/3 + \ldots\ldots n^n/n$**

Aim To calculate the sum of the series : $1^1/1 + 2^2/2 + 3^3/3 + \ldots\ldots n^n/n$

Coding:

```
n = int(input("Enter a value of n: "))
s = 0
for i in range(1,n+1):
    a= (i**i)/i
    s=s+a
print("The sum of the series is ", s)
```

Output:

```
Enter a value of n: 4
The sum of the series is  76
```


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